Deep Learning Pipeline for Liver Vessel Segmentation

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ABSTRACT

This capstone project involves the development and deployment of a comprehensive deep learning pipeline for segmenting liver vessels from triple-phase liver CT scans, aimed at aiding surgical planning for liver transplants and enhancing radiological assessments. Set against the rapidly evolving landscape of Artificial Intelligence (AI) in medical imaging, this project introduces an end-to-end application tailored for clinical deployment at Cleveland Clinic Abu Dhabi (CCAD).

The project was divided into two phases. The first phase focused on the rapid development of a deep learning model using Convolutional Neural Networks (CNNs) to accurately segment liver vessels. The second phase concentrated on deploying this model in a clinical setting, ensuring its integration with existing healthcare systems and evaluating its performance and usability.

Key challenges included data preprocessing, training the deep learning model, and effective deployment. The model achieved an average DICE coefficient of approximately 0.71, demonstrating its feasibility for clinical use, especially for liver transplants where accurate segmentation of the main vessel branches is critical. The deployment phase validated the model's integration with hospital PACS systems, showcasing the practical applicability of this AI-driven solution.

Future directions include expanding the dataset, enhancing model accuracy through advanced techniques, developing interactive interfaces for real-time clinician feedback, and reducing the need for triple-phase CT scans to dual-phase CT scans, thus minimizing radiation exposure. This project highlights the potential of deep learning in medical imaging, particularly in the context of liver transplants, and sets the stage for continued improvements and broader clinical adoption.

KEYWORDS

Liver Vessel Segmentation, Deep learning, Triple-phase liver CT scans, Convolutional Neural Networks, Medical Image Segmentation, MONAI, AI in Radiology

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1 INTRODUCTION

The primary motivation for this capstone project, being developed in collaboration with Cleveland Clinic Abu Dhabi (CCAD), was to address a critical challenge in liver transplantation: the precise mapping of liver vasculature for surgical planning. This project developed a deep learning pipeline for liver vessel segmentation from triple-phase liver CT scans, a task essential for successful liver transplant surgeries.

Liver transplantation demands high accuracy in preoperative planning, where detailed mapping of the liver's vascular structure is crucial. Current methods for liver vessel segmentation often lack the necessary precision and efficiency, leading to potential risks in surgical planning. This project aims to overcome these limitations by employing deep learning techniques in medical imaging, specifically tailored for the segmentation of liver vessels.

The focus of this project extended beyond the development of a segmentation model. A significant aspect involves the integration of this model into the existing healthcare infrastructure. While advancements in AI bring forward new networks with improved segmentation accuracies, their practical application in clinical settings is not widespread. This project addressed this gap by creating a software solution that not only incorporates state-of-the-art deep learning techniques but also aligned with the operational workflows of healthcare professionals.

There are two challenges in this project: technical and practical. Technically, the project aims to deal with the variability in liver vessel structures and manage real volumes

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of data from triple-phase CT scans. It also required continual updates to integrate emerging segmentation techniques. Practically, the deployment in a clinical setting required integration with existing medical systems, intuitive user interface design, and usability for healthcare professionals.

This paper details the methodology adopted in developing the deep learning pipeline, the challenges encountered, and the strategies employed to address them. It is also in the process of presenting the initial deployment of this system at CCAD. Through this project, we aim to improve the accuracy of liver vessel segmentation and its integration into medical practice, ultimately enhancing the quality of surgical planning and patient outcomes in liver transplantation.

2 RELATED WORK

2.1 Introduction

Segmentation in liver imaging, a crucial process in medical imaging, isolates liver vessels for diagnostic and treatment planning. This is challenging due to the intricate and fine structure of liver vessels, their variability among individuals, and the complexity involved in differentiating these vessels from the surrounding tissues[9].

A key advancement that has facilitated the distinction of liver vessels in imaging is the Triple Phase Liver protocol[3]. This medical imaging technique involves acquiring CT scans during three phases of contrast enhancement: the arterial, venous, and delayed phases. Each phase offers a unique contrast profile, enhancing the visibility of liver vessels against the surrounding tissue. This enhanced contrast is crucial in making liver vessels more distinguishable for accurate segmentation[9].

The importance of precise liver vessel segmentation plays a pivotal role in various medical applications such as surgical planning, where it is essential to avoid damaging critical vessels. Accurate mapping of these vessels is also crucial in the diagnosis of liver diseases and treatments like liver transplants. As such, the advancements in liver vessel segmentation techniques present a significant stride in improving patient outcomes in liver-related healthcare.

2.2 Current Methods and Challenges

A comprehensive review of computational methods for liver vessel segmentation in medical imaging [3] classifies these techniques into three main categories: image-based, model-based, and machine learning-based methods. Image-based methods focus on analyzing image properties, such as intensity, gradient, and texture, while model-based methods rely on prior knowledge of the liver vessel structure to perform segmentation [3]. Traditional approaches, including image filtering and enhancement, deformable models, tracking, and early machine learning-based methods, often face limitations such as the need for parameter adjustments, sensitivity to initial conditions, and user interaction, which can negatively impact their accuracy and efficiency in clinical settings [3].

Additionally, these methods may struggle with variations in liver vessel structures and the presence of noise in medical images, leading to inaccurate segmentation results [3]. Furthermore, the complexity of liver vessel structures, the presence of branching patterns, and the overlap of vessels with other anatomical structures pose challenges to traditional segmentation techniques [3].

The growing popularity of deep learning techniques in liver vessel segmentation is due to their ability to extract relevant features from images without requiring specialist knowledge. However, these methods largely depend on the availability of large and labeled datasets, which are still limited. Recent trends in deep learning include unsupervised and semi-supervised learning, as well as generative networks, but their application in liver vessel segmentation is still limited [3]. To develop reliable and clinically useful liver vessel segmentation methods, there is a need for more publicly available datasets, standardized evaluation metrics, and close collaboration between experts in medicine, software engineering, and statistical analysis [3].

The advent of deep learning has led to significant advancements in medical image segmentation [6]. Deep learning techniques, particularly convolutional neural networks (CNNs), have demonstrated their ability to automatically learn complex features and achieve state-of-the-art performance in a wide range of medical imaging tasks [6]. Despite their success, deep learning methods for liver vessel segmentation face challenges such as limited training samples, unbalanced class frequency, and incomplete or inaccurate annotations in benchmark datasets [6, 7]. To address these challenges, researchers have proposed various techniques for data preprocessing, augmentation, and post-processing to improve the performance of deep learning models in liver vessel segmentation tasks [6].

Furthermore, deep learning techniques require large amounts of computational resources and can suffer from overfitting when trained on small datasets [6]. To overcome these limitations, researchers have proposed transfer learning, which involves pre-training models on large datasets and fine-tuning them on the target task, as well as employing ensemble techniques to combine multiple models for improved performance [6]. Despite these advancements, there is still room for improvement in the application of deep learning techniques for liver vessel segmentation to ensure accurate and efficient results in clinical settings.

2.3 Advancements in Liver Vessel Segmentation

In terms of deep learning breakthroughs, the field continues to innovate rapidly. The U-Net architecture, with its encoder-decoder structure, has become particularly influential in medical image segmentation. Its design is adept at capturing both contextual information and fine details, making it highly suitable for tasks such as liver vessel segmentation. Similarly, the ResNet model, known for introducing residual learning, has enabled the training of deeper networks, thereby enhancing the capability to tackle more complex segmentation tasks. CNN architectures like U-Net, ResNet50, and 3D-Res-UNet have shown promise in liver segmentation. U-Net, specifically designed for biomedical imaging, excels in accuracy and efficiency, while ResNet50 is notable for its deep layers and residual connections. 3D-Res-UNet combines U-Net's feature extraction with ResNet's gradient flow, showing significant results in addressing noise, low contrast, and varying vessel structures [2, 11, 13].

A novel approach using 3D U-Net with a variant dice loss function has been proposed for robust liver vessel extraction, addressing challenges like limited training samples and unbalanced class frequency [7]. This method effectively segments liver vessels, overcoming issues with noise, contrast, and varied structures. The use of a variant dice loss function and parameter adjustments enhances segmentation accuracy.

Further advancements have been seen in the adaptation and modification of these networks. For instance, incorporating attention mechanisms, as seen in Attention U-Net, and employing multi-scale approaches, as in H-DenseUNet, have significantly improved the segmentation of fine vessel structures. The field is also witnessing innovations like the integration of Transformer models and the use of generative adversarial networks (GANs) for generating synthetic training data, which are contributing to enhanced model robustness.

A noteworthy success in applying CNNs to medical image segmentation is the Total Segmentator. This tool utilizes deep learning techniques for the automatic segmentation of over 104 anatomical structures from CT images. It represents a significant leap in automating a traditionally labor-intensive process, providing rapid and precise segmentations. Total Segmentator's design caters to the complexities and variability inherent in medical imaging data, adapting to different scan qualities and anatomical variations. Its effectiveness in clinical settings and research underscores the practical benefits of CNNs and validates the extensive research investment in this technology[14].

2.4 Radiology Workflows and AI Integration

AI-powered segmentation tools have shown remarkable precision in identifying and delineating anatomical structures, leading to more accurate diagnoses. The rapid results provided by these tools have shortened turnaround times for radiological reports, crucial for timely clinical decision-making. In the specific context of liver vessel segmentation, the MONAI framework, a PyTorch-based platform, stands out. MONAI, and its component MONAI Label, specifically addresses the scarcity of annotated datasets for supervised learning in medical imaging [1, 4]. The application of MONAI Label in the Telestroke Service at Prince of Wales Hospital for brain hemorrhage segmentation highlights its effectiveness in medical imaging. It demonstrates the framework's adaptability to specific tasks, such as liver vessel segmentation, thereby streamlining development and enhancing the efficiency of AI applications in radiological workflows.

2.5 DICOM Standards in Medical Imaging

The Digital Imaging and Communications in Medicine (DICOM) standard plays a central role in the integration of AI into medical imaging[10]. As the universal protocol for handling, storing, and transmitting medical imaging information, DICOM's compatibility with AI applications is crucial. This compatibility ensures that new AI-driven technologies can be smoothly incorporated into existing radiological practices without disrupting established systems and protocols. Open-source software like OHIF (Open Health Imaging Foundation) exemplifies how platforms can support the management of DICOM images while integrating AI tools for advanced functionalities[5].

2.6 Gaps and Opportunities

The review of related works in liver vessel segmentation and the integration of AI in medical imaging has illuminated critical gaps and emerging opportunities in the field. Despite significant advancements in deep learning and AI technologies, challenges persist, particularly regarding limited datasets and the need for accurate annotations. This capstone project aims to address these gaps by exploring innovative data augmentation and model training techniques, contributing to enhanced segmentation accuracy and efficiency. Additionally, the integration of AI into radiological workflows, leveraging platforms like MONAI and MonaiLabel, offers an opportunity to streamline diagnostic processes and improve clinical outcomes.

Furthermore, the deployment of AI in clinical settings, while challenging, presents a substantial opportunity for advancing patient care. This project aligns with these needs by developing AI tools that are robust, accurate, and compliant

with healthcare regulations, thus enhancing their viability in clinical environments. The insights from this literature review establish a solid foundation for the project, highlighting its potential to make significant contributions to medical imaging and AI, particularly in refining liver vessel segmentation techniques. As the project progresses, these insights will guide the development of solutions that are both technologically innovative and practically applicable in healthcare settings.

3 METHODOLOGY

3.1 Objective

3.1.1 Phase 1. The primary objective of Phase 1 of this capstone project was to develop a deep learning model for the segmentation of liver vessels from triple-phase liver CT scans. This phase laid the groundwork for an end-to-end radiology application that leveraged machine learning techniques, like Convolutional Neural Networks (CNNs), to segment the imaging data. The goals included:

- **Data Preprocessing and Analysis:** This involved the preprocessing of training data, specifically triple-phase liver CT scans, preparing them for deep learning model training.
- **Rapid Deep Learning Model Development:** This included the development of a CNN model tailored for liver vessel segmentation. This process included the selection, training, and optimization of CNN networks to enhance their accuracy and adaptability in segmenting liver vessel structures.

The success of Phase 1 was pivotal in demonstrating the feasibility and effectiveness of AI applications in medical imaging, particularly in the challenging task of liver vessel segmentation.

3.1.2 Phase 2. Looking ahead to Phase 2, the project expanded upon the foundation established in the first phase. The objectives included:

- **Broader Deployment and Integration:** This phase focused on trying out different iterations of the model for deployment. This included ensuring the application's integration into radiological workflows and evaluating its performance in clinical settings.
- **Orchestration:** The deployment strategy also involved testing and using infrastructure that managed routing, processing (with the previously built model as a module), and scheduling. This allowed for seamless connection and integration with different clinical PACS systems.

The overarching goal of this two-phase project is not only to develop a tool for liver vessel segmentation but also to create a software solution that is practical, efficient, and beneficial in the realm of medical diagnostics and treatment planning.

3.2 Data Acquisition and Characteristics

The dataset utilized in this project consisted of triple-phase CT scans from 26 subjects, obtained from Cleveland Clinic Abu Dhabi. Triple-phase CT scanning is a medical imaging technique that captures three distinct image sets of the liver, each corresponding to a different phase of blood flow: hepatic arterial, portal venous, and late venous phases. This method was crucial for delineating the liver's vascular structure, which is essential for the segmentation tasks in this project. A 2D representation of each of the three phases is shown in Figure 1.

The images are stored in the DICOM format, widely recognized in medical imaging for its standardization in handling, storing, and transmitting image data. The DICOM metadata for our data includes specific details such as the type of contrast used (e.g., Optiray 350), the scanner type (e.g., SIEMENS SOMATOM Definition Flash), and technical specifications like KVP, pixel spacing, and exposure time. This information is crucial for understanding the imaging context, including the type of sequence used and the oral contrast details. The volumes are of size 420x340x190.

Accompanying the imaging data are vessel labels in Rtstruct format, typically used in radiation therapy for defining anatomical structures and crucial for training and evaluating the deep learning model.

3.3 Data Preparation and Preprocessing

Following dataset acquisition, the initial step in our methodology involved preprocessing the acquired data. This crucial phase entailed converting both the DICOM images and the accompanying Rtstruct labels into the Nifti format. The Nifti (Neuroimaging Informatics Technology Initiative) format is widely used in the field of medical imaging due to its support for three-dimensional data and compatibility with various AI models and analytical tools.

3.3.1 Initial Processing and Conversion. The conversion of DICOM images to Nifti was a straightforward process that is easily done by libraries such as dcm2nii. However, converting the Rtstruct labels, which are typically used in radiation therapy for labelling anatomical structures, posed a challenge. These labels are embedded in complex file structures that are not natively compatible with standard AI model processing pipelines.

To address this, we developed a specialized conversion pipeline using MATLAB. This custom pipeline was designed to navigate the complexity of the Rtstruct files, ensuring that the structural information they contained was accurately translated into the Nifti format.

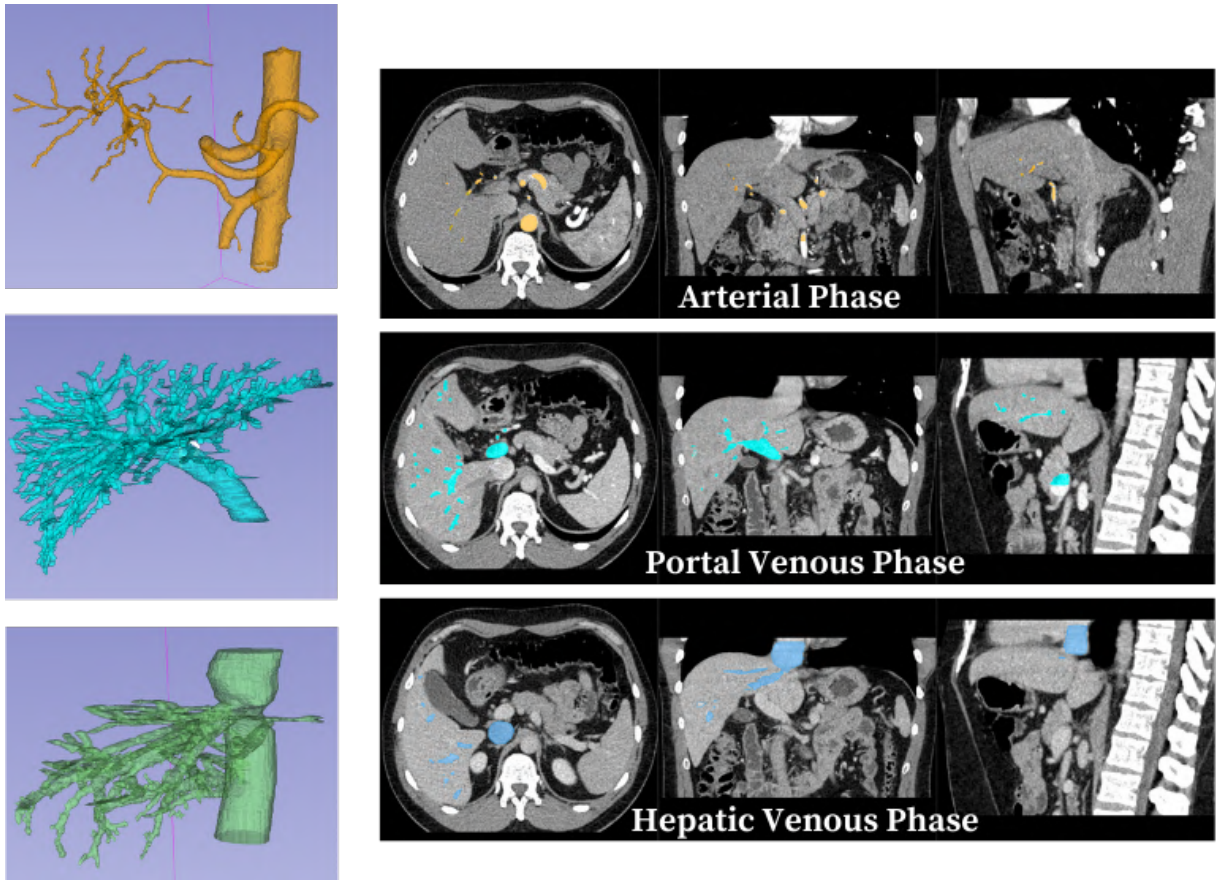


Figure 1: A triple phase CT scan along with segmented vessels with 3D visualized labels

3.3.2 Issue with Conversion Pipeline. During conversion, we noticed a loss of critical metadata, including pixel intensity information and image orientation. Preserving this metadata is vital for accurate image analysis, influencing model training and segmentation accuracy.

3.3.3 Resolution and Testing. We revised the conversion pipeline to retain all essential metadata. The resulting Nifti files included detailed metadata, such as dimensions -424x346 x193), pixel spacing (0.79x0.79x1.00), and orientation details, ensuring the fidelity of the imaging data.

With the metadata integrity ensured, the preprocessed data in Nifti format was ready for subsequent phases. This data was then tested using the MONAI framework, setting the foundation for model development.

3.4 Data Augmentation and Transformations

Given the size of our dataset, data augmentation was a crucial aspect of our model training. It was implemented through on-the-fly transformations. This method involves dynamically altering the data each time it is accessed during training, thereby effectively expanding the diversity of the dataset without physically increasing its size.

The implementation of data augmentation, including a variety of transformations, was executed through a predefined series of steps within our training pipeline:

- **Label Normalization and Spacing**: The labels are normalized for consistency, while the spacing of the images is standardized for uniform scale. This step also includes cropping to focus on relevant regions, thereby reducing computational load.
- **Spatial Padding and Intensity Scaling**: Spatial padding ensures uniform spatial size across images and labels, aligning with the desired ROI. Intensity scaling adjusts pixel values, normalizing and enhancing image features for better visibility and analysis.
- **Random Cropping by Labels**: This augmentation randomly crops the images based on the presence of labels. This step is critical for training the model to identify both the presence and absence of liver vessels.

- **Random Flipping along Axes**: The images were randomly flipped along different axes.
- **Rotation and Intensity Shifting**: Random 90-degree rotations and shifts the intensity of the images were applied randomly.

3.4.1 Impact of On-the-Fly Augmentation: This on-the-fly augmentation approach is beneficial as it virtually expands the dataset size, continually generating new variations based on the number of training epochs. Such a method is particularly effective in preventing overfitting and ensuring the model's ability to generalize well to new, unseen data.

The outlined data augmentation strategy ensures that the model is exposed to a diverse range of anatomical variations, which is important for its generalizability in real-world medical imaging applications.

3.5 Evaluation Metric

For assessing the model's performance, we employed the DICE coefficient as the primary evaluation metric. Mathematically, the DICE coefficient is formulated as follows:

$$\text{DICE} = \frac{2 \times |X \cap Y|}{|X| + |Y|}$$

This metric quantifies the overlap between the predicted segmentation and the ground truth, providing a robust measure of accuracy. In the context of liver vessel segmentation, where precision is important due to the complexity of the anatomical structures, the DICE coefficient offers a reliable assessment of the model's performance. It ensures that the model not only captures the primary hepatic vessels accurately but also identifies the finer, more intricate vessels effectively.

3.6 Testing and Validation

3.6.1 Dataset Partitioning. For model evaluation, the dataset comprising triple-phase CT scans from 26 subjects was partitioned, allocating 2 subjects exclusively for testing purposes. The remaining 24 subjects' data were used for model training. This partitioning strategy ensures that the model is tested on unseen data.

3.6.2 Validation During Training. During the training phase, a dynamic validation approach was employed, leveraging MONAILabel framework's on-the-fly data transformations. Unlike static validation techniques such as K-fold cross-validation, this method dynamically loads the validation dataset the same way it loads the training dataset. This continuous refreshing of the validation set helped prevent overfitting and ensured the model's reliable performance on new, unseen data.

3.7 Hyperparameters

The optimization of hyperparameters was a critical component of the training process, conducted through extensive initial testing. Key parameters were meticulously selected to achieve an equilibrium between training efficacy and computational efficiency:

- **Target Spacing**: Target spacing refers to the physical space between data points in the CT scans. For our model, we chose a target spacing of 0.8mm in all three dimensions (x, y, and z). This spacing allows the model to process the scans at a resolution that is detailed enough to capture essential features while being computationally manageable.
- **Learning Rate and Weight Decay**: The AdamW optimizer, a modification of the standard Adam optimizer, was employed with a learning rate of 1e-4 and a weight decay of 1e-5. Optimizers in machine learning are algorithms used to adjust network parameters to minimize loss functions. AdamW enhances Adam by separately handling weight decay, improving training stability and preventing overfitting. The chosen learning rate ensures gradual and effective model weight adjustments, while the weight decay helps in regulating the model's complexity.

3.8 Network Architectures Used

A series of experiments were conducted to evaluate various CNN architectures for liver vessel segmentation. Each network was assessed based on its structural design, including the number of layers, strides, and other hyperparameters. These architectural choices play a pivotal role in determining the network's ability to accurately segment complex anatomical structures like liver vessels.

Each of the CNNs tested below were trained for 500 epochs with an ROI size of (256, 256, 128), except for the UNETR for which an ROI of (128, 128, 128) was used due to computational limitations.

3.8.1 UNet.

- **Architecture**: UNet's design features an encoder-decoder structure optimized for biomedical image segmentation. The encoder progressively reduces spatial dimensions while increasing feature channels, capturing context. The decoder then gradually recovers object details and spatial dimensions. UNet's symmetric structure and skip connections facilitate precise localization.
- **Hyperparameters**:
  - Channels: [16, 32, 64, 128, 256]
  - Strides: [2, 2, 2, 2]
  - Number of Residual Units: 2

- Normalization: Batch

3.8.2 *SegResNet.*

- **Architecture**: SegResNet integrates segmentation and residual learning, employing blocks that consist of convolutional layers with residual connections. These connections help mitigate the vanishing gradient problem, enabling the training of deeper networks.
- **Hyperparameters**:
  - Initial Filters: 32
  - Blocks Down: (1, 2, 2, 4)
  - Blocks Up: (1, 1, 1)
  - Dropout Probability: 0.2

3.8.3 *UNETR.*

- **Architecture**:UNETR adapts the Transformer architecture for medical imaging, integrating self-attention mechanisms for segmentation tasks. It uses a combination of convolutional and Transformer blocks to process 3D medical images. This structure allows the network to capture both local features through convolutional layers and global dependencies via self-attention.
- **Hyperparameters**:
  - Feature Size: 64
  - Hidden Size: 1536
  - MLP Dimension: 3072
  - Number of Heads: 48
  - Position Embedding: "conv"
  - Normalization: "instance"
  - Residual Blocks: Enabled

3.8.4 *DynUNet.*

- **Architecture**: DynUNet is a dynamic convolutional network that adapts its structure based on the input data. It employs a series of convolutional layers with varying kernel sizes and strides, allowing the network to learn from a diverse range of feature scales. The inclusion of instance normalization and optional deep supervision enhances the model's ability to focus on relevant features.
- **Hyperparameters**:
  - Kernel Size: [3, 3, 3, 3, 3, 3]
  - Strides: [1, 2, 2, 2, 2, [2, 2, 1]]
  - Upsample Kernel Size: [2, 2, 2, 2, [2, 2, 1]]
  - Normalization: "instance"
  - Residual Blocks: Enabled.

3.9 Results

Table 1 summarizes the performance metrics of various CNN architectures tested in the previous section, focusing on the portal venous phase of liver vessel segmentation. Each architecture demonstrates its unique strengths in segmentation accuracy, as reflected in their respective DICE coefficients.

Network	Best DICE Coefficient
UNet	0.7514
SegResNet	0.7519
UNETR	0.7452
DynUNet	0.7563

Table 1: Performance Metrics of CNN Architectures for Liver Vessel Segmentation (Portal Venous Phase)

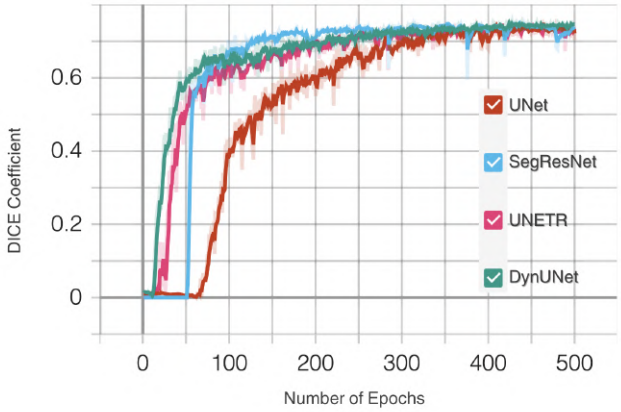


Figure 2: Validation DICE Score for different networks

The models proficiencies were notable in accurately capturing the primary hepatic vessels and their initial bifurcations, as shown in Figure 3. However, it encountered difficulties in segmenting smaller, more intricate vessels towards the end of the hepatic vasculature.

The model trained using UNet was selected for ongoing development due to its proven effectiveness in medical image segmentation. Its performance, demonstrated in our trials, and in other medical imaging tasks, makes it a reliable choice.

Phase	DICE Score
Hepatic Artery	0.67
Portal Venous	0.75
Hepatic Venous	0.70

Table 2: Triple Phases performance with UNet

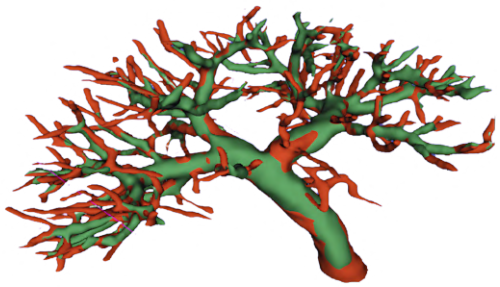


Figure 3: Predicted (Green) and Ground Truth (Red) for Liver Vasculature

3.10 Further Experiments and Observations

Additional attempts were made in an attempt to gain insights and improve segmentation accuracy.

3.10.1 Localizing the Liver for Targeted Training. TotalSegmentator, an automated organ segmentation tool, was employed to generate accurate liver labels, facilitating these targeted training methods.

Two approaches were explored:

(1) **Approach 1 - Liver Cropping:** The first method involved localizing and cropping the liver using a liver mask. This approach primarily enhanced the training speed, allowing the model to converge more quickly.

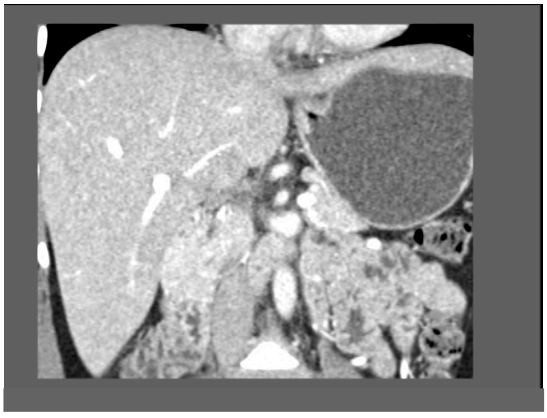


Figure 4: Grey zone depicting part of volume that is cropped out while training

(2) **Approach 2 - Masking the Liver Volume:** The second method masked the entire volume, focusing the training exclusively on the vessels within the liver. This resulted in a decrease in segmentation accuracy, particularly affecting the detection of main branches outside the liver, explaining the drop in performance.

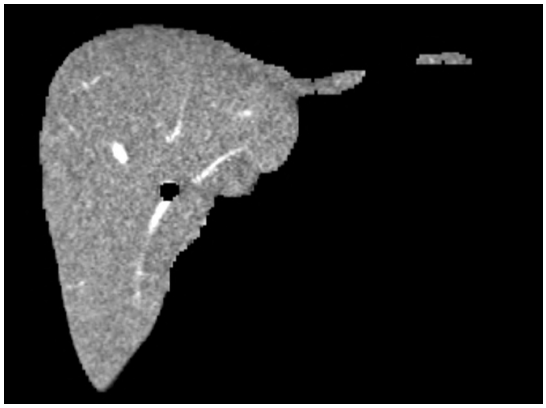


Figure 5: A mask of the liver showing part of volume getting trained

These attempts to refine segmentation by localizing the liver and modifying training focus did not significantly improve the model's ability to segment finer vessels. The results indicate that maintaining the same training configurations, even with localized training, may not be the most effective strategy for detailed hepatic vasculature segmentation.

3.10.2 Impact of Dataset Size on Model Performance. In an effort to understand the impact of dataset size on the performance of our liver vessel segmentation model, we conducted a strategic experiment involving the intentional reduction of the dataset. The primary objective of this experiment was to determine if there exists a direct correlation between the size of the dataset and the model's effectiveness in segmentation, as quantitatively measured by the DICE score.

It was observed that a decrease in the dataset size consistently led to a reduction in the DICE score. This reduction was quantifiable and consistent, providing clear evidence of the relationship between dataset size and model performance. Specifically, we noted that the removal of every two volumes from the dataset resulted in an approximate 0.04 decrease in the DICE score. This finding was significant as it quantitatively demonstrated the sensitivity of the model's performance to the volume of training data.

4 MODEL DEPLOYMENT

After completing the development phase, the focus shifted to the second objective: testing various deployment methods to evaluate the model's performance and usability in real-world clinical settings. This step was crucial for assessing the

integration of the model into clinical workflows and its adaptability to the needs of medical professionals. Insights gained from these deployments were invaluable for informing the continued development and optimization of the model.

The following iterations were developed during model deployment:

(1) Development with 3D Slicer and MONAI Label
(2) Flask Web Application
(3) noGUI Shell Application
(4) Integrated DICOM Web Viewer and DICOMWeb Server

4.0.1 Initial Development with 3D Slicer and MONAI Label. The initial setup involved integrating the deep learning models with 3D Slicer and the MONAI Label server. 3D Slicer, a free, open-source platform for medical image informatics, image processing, and 3D visualization, functioned as the local client. It provided a user-friendly interface for visualizing and interacting with the segmentation results produced by our models. The MONAI Label server, running on a remote setup, hosted the deep learning models, facilitating the processing and analysis of medical images. This phase tested the model's functionality, and was also the setup used during development.

4.0.2 Flask Web Application. The first iteration of deployment was a Flask web application designed for online deployment. This application allows users to upload CT scan volumes and receive segmented labels in return. Interfacing with the MONAI Label server's APIs, the Flask application served as a foundational step in understanding the dynamics of a segmentation-based web application. Its development is similar to the online deployment model of TotalSegmentator, providing insights into the potential of web-based deployments for public accessibility and use.

4.0.3 Development of a NoGUI Shell Application. To cater to environments where a graphical user interface (GUI) is not preferable, a NoGUI application was developed. This application executed the model via shell commands, processing designated volume paths and outputting the results along with a basic report. The report included 2D representations of the volume slices with labels and quantified the liver vessels' volumes. This step was instrumental in exploring automated, streamlined processing methods suitable for a setting where batch processing was required.

4.1 Orchestration and Integration with DICOM Standards

A significant aspect of the deployment phase involved utilizing Mercure, an open-source DICOM orchestration platform, to manage the routing, processing, and scheduling of DICOM studies. This section provides an in-depth overview of Mercure's features, their importance, and the role of DICOM standards in this integration, particularly focusing on their relevance to our project.

4.2 Mercure Overview

Mercure is a flexible and intuitive platform designed to facilitate the orchestration of DICOM workflows. It supports the dispatching of DICOM studies to various targets based on definable routing rules and the processing of DICOM series using custom-developed algorithms. Mercure offers a web-based user interface and extensive monitoring options, making it suitable for routine applications that require high availability.

In our project, Mercure was instrumental in managing and orchestrating the complex workflow of processing liver vessel segmentation tasks. The integration of Mercure allowed for seamless data flow and ensured that each DICOM study was efficiently processed and routed to the appropriate module and PAC system.

4.3 Key Features and Their Importance

- **Routing and Processing**: Mercure can route DICOM studies to different targets and process them with custom algorithms, such as our AI models for liver vessel segmentation. These processing algorithms run as Docker containers on the Mercure server or connected cluster nodes. This feature was crucial for our project as it ensured the correct processing of DICOM studies without manual intervention, thereby streamlining the workflow.
- **Integration with MONAI**: Mercure supports MONAI Application Packages (MAP), allowing seamless integration of AI models developed with the MONAI framework. This compatibility enabled us to directly incorporate our liver vessel segmentation models into the Mercure orchestration, enhancing their usability and performance in clinical settings.
- **Scalability and Flexibility**: The platform can scale services and handle complex workflows, making it suitable for large-scale clinical deployments. For our project, this meant that as the volume of DICOM studies increased, Mercure could efficiently manage the load, ensuring consistent performance and reliability.
- **Efficient Workflow Management**: By automating the routing and processing of DICOM studies, Mercure ensured that the right data reached the appropriate processing modules without manual intervention. This efficiency was vital in maintaining the workflow's smooth operation, reducing potential delays in processing.

4.4 Integration with DICOM Standards

Adhering to DICOM standards was crucial for the clinical deployment of our model. DICOM (Digital Imaging and Communications in Medicine) is the universal standard for handling, storing, and transmitting medical imaging information. Ensuring compatibility with DICOM standards facilitated seamless integration with existing medical imaging technologies and radiological workflows.

Key aspects of DICOM integration included:

- **Orthanc and OHIF Viewer**: We used Orthanc, a lightweight RESTful DICOM server, and the OHIF Viewer to enhance radiologists' interaction with processed medical images. Orthanc served as the backend for storing and managing DICOM files, while the OHIF Viewer provided a web-based interface for viewing and annotating these images.
- **Compliance and Interoperability**: Ensuring that our model's outputs were compliant with DICOM standards guaranteed interoperability with other systems in the hospital. This was essential for smooth integration into the clinical workflow and for the practical utility of the application.

4.4.1 Implementation and Containerization. The entire system, including the model, MONAI Label server, and DICOM tools, was containerized using Docker. This approach ensured scalability and facilitated easy adoption by hospitals. Hospitals could connect to the system through their existing web viewers and servers, allowing for streamlined integration into their clinical processes. The use of the MONAI Application package ensured the integration and functionality of the deep learning model within the clinical workflow.

5 FUTURE DIRECTIONS

As our project progresses, several key areas have been identified for further development to enhance the effectiveness and applicability of our deep learning pipeline for liver vessel segmentation. These future directions focus on expanding datasets, enhancing model accuracy, and developing interactive interfaces for clinical use.

5.0.1 Dataset Expansion. One of the critical next steps is the expansion of our dataset. Acquiring triple-phase CT scans is challenging due to the invasive nature of the procedure. To address this, we plan to gather more diverse datasets from various healthcare facilities. Additionally, we will explore generating synthetic datasets using advanced techniques like diffusion models. This approach will help to increase the volume and diversity of training data, which is crucial for improving the model's robustness and generalizability across different patient populations and pathological conditions.

5.0.2 Enhancing Model Accuracy. To better handle the complexities of liver vascular structures, we will explore ensemble methods and other advanced machine learning strategies. By combining the predictions of multiple models, we aim to enhance overall segmentation accuracy. This approach will involve developing models that specialize in different aspects of liver vessel segmentation, such as capturing the overall structure versus detailing finer vascular features. Furthermore, integrating attention mechanisms and advanced techniques like Generative Adversarial Networks (GANs) will provide new ways of capturing intricate vessel structures, leading to a more nuanced and accurate representation.

5.0.3 Interactive Interface Development. Creating a user friendly interface for active learning is another crucial future direction. Such an interface will allow clinicians to provide real-time feedback, enhancing the model's adaptability and customization to clinical needs. By incorporating Reinforcement Learning with Human Feedback (RLHF), we can establish a collaborative feedback loop where radiologists' inputs directly influence the model training process. This interactive approach will not only improve the model's accuracy but also ensure its practical utility and ease of use in clinical settings.

5.0.4 Directions for Deployment. A crucial aspect of future deployment involves creating a specialized application for Cleveland Clinic Abu Dhabi (CCAD). This application aims to replace the existing liver segmentation methodology with our locally developed AI application. The integration will streamline and enhance the surgical planning process, crucial for successful transplant surgeries. The app will focus on user-friendliness and compatibility with existing healthcare IT infrastructure, ensuring it becomes an integral part of the surgical planning process.

5.0.5 Replace Triple-phase with Dual-phase. Another significant future direction, once the model's segmentation quality is sufficiently high, is to replace the need for a triple-phase CT scan with a dual-phase CT scan. This reduction in the number of required phases will decrease radiation exposure by one-third, enhancing patient safety. Additionally, there is potential to supplement the replaced phase with MRI scans, offering a non-invasive alternative that further minimizes radiation exposure. This advancement could significantly improve the clinical workflow and patient experience in liver vessel imaging.

6 CONCLUSION

This project aimed to develop a deep learning pipeline for liver vessel segmentation to assist in surgical planning for liver transplants. The project was successful in building and

deploying this pipeline, demonstrating its viability in a clinical setting. The deep learning model achieved a DICE coefficient of approximately 0.71. While this accuracy might seem low, it is important to note that the model struggled primarily with the segmentation of finer vessel structures at the distal ends of the vasculature. However, for liver transplants, the critical requirement is the accurate segmentation of the main branches of the liver vessels, which our model achieved effectively. Consequently, the model's performance is sufficient for its intended clinical use, where it can serve as a valuable assistance tool for surgeons.

The deployment phase successfully demonstrated the integration of the model with hospital PACS systems. By using tools like Mercure for orchestration and adhering to DICOM standards, we ensured seamless data flow and efficient processing within the hospital environment. This validated the practical applicability of our AI-driven solution in a real-world clinical setting. Overall, this project has shown that an AI-driven solution for liver vessel segmentation is viable and can significantly aid in surgical planning for liver transplants. The project met its primary objectives, laying a strong foundation for future enhancements. Despite the successes, there is considerable room for improvement. Future directions include expanding the dataset, enhancing model accuracy through advanced techniques, and developing interactive interfaces for real-time clinician feedback. A high-impact future goal is to reduce the need for triple-phase CT scans to dual-phase CT scans, thereby minimizing radiation exposure. In conclusion, this project has demonstrated the potential of deep learning in medical imaging, particularly in the context of liver transplants. With continued improvements and expansions, this AI-driven solution has the potential to significantly enhance clinical workflows and patient outcomes.

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